Basic Concepts

Software Architecture
Lecture 3

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What is Software Architecture?

- **Definition:**
  - A software system's architecture is the set of *principal design decisions* about the system
- Software architecture is the blueprint for a software system's construction and evolution
- Design decisions encompass every facet of the system under development
  - Structure
  - Behavior
  - Interaction
  - Non-functional properties

Design decisions encompass every aspect of the system under development, including:

- Design decisions related to system *structure*—for example, "The architectural elements should be organized and composed exactly like this . . ."
- Design decisions related to *functional behavior*—for example, "Data processing, storage, and visualization will be performed in strict sequence."
- Design decisions related to *interaction*—for example, "Communication among all system elements will occur only using event notifications."
- Design decisions related to the system's *nonfunctional properties*—for example, "The system's dependability will be ensured by replicated processing modules."
- Design decisions related to the system's *implementation*—for example, "The user interface components will be built using the Java Swing toolkit."

What is "Principal"?

- "Principal" implies a degree of importance that grants a design decision "architectural status"
  - It implies that not all design decisions are architectural
  - That is, they do not necessarily impact a system's architecture
- How one defines "principal" will depend on what the stakeholders define as the system goals

Other Definitions of Software Architecture

- Perry and Wolf
  - Software Architecture = { Elements, Form, Rationale }
    what *how* *why*
- Shaw and Garlan
  - Software architecture [is a level of design that] involves
    - the description of elements from which systems are built,
    - interactions among those elements,
    - patterns that guide their composition, and
    - constraints on these patterns.
- Kruchten
  - Software architecture deals with the design and implementation of the high-level structure of software.
  - Architecture deals with abstraction, decomposition, composition, style, and *aesthetics*.

Temporal Aspect

- Design decisions are and unmade over a system's lifetime
  - ☐ Architecture has a temporal aspect
- At any given point in time the system has only one architecture
- A system's architecture will change over time

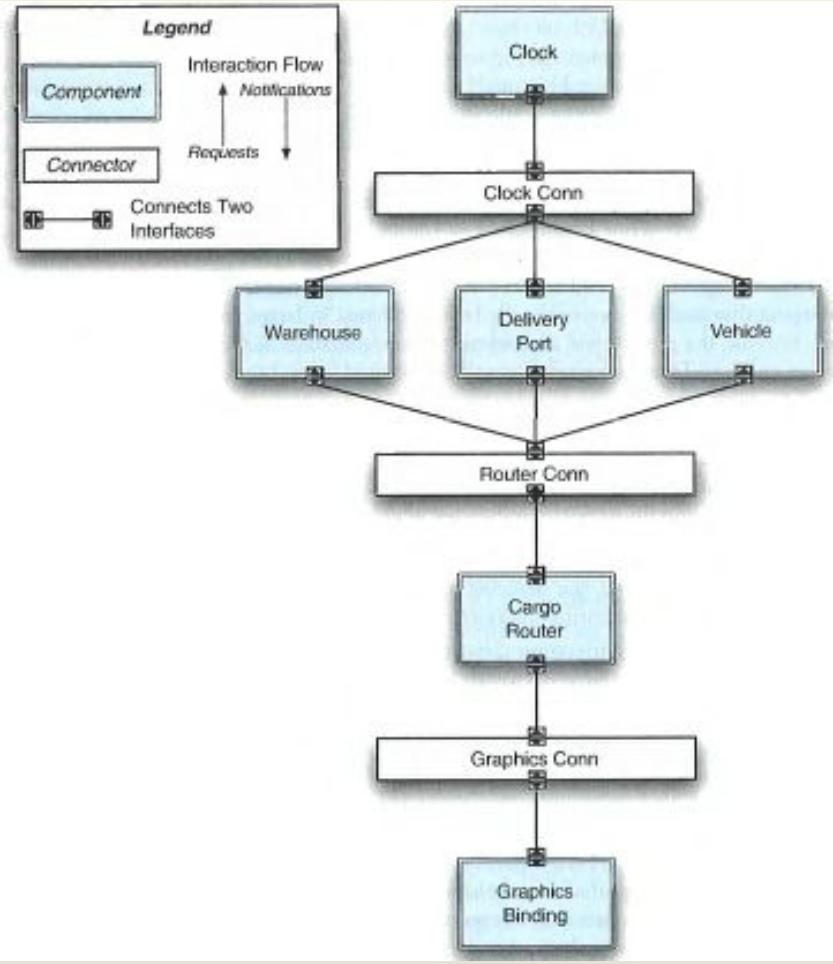
Prescriptive vs. Descriptive Architecture

- A system's *prescriptive architecture* captures the design decisions made prior to the system's construction
  - It is the *as-conceived* or*as-intended* architecture
- A system's *descriptive architecture* describes how the system has been built
  - It is the *as-implemented* or *as-realized* architecture

An Example

This application controls the routing of cargo from a set of incoming Delivery Ports to a set of Warehouses via a set of Vehicles. The Cargo Router component tries to optimize the use of vehicles and deliver the cargo to its proper destination (that is, warehouse). The Clock component provides the time and helps the ports, vehicles, and warehouses to synchronize as necessary. Finally, the Graphics Binding component provides a graphical user interface that allows a human operator to assess the state of the system during its execution. The components in the system interact via three connectors—Clock Conn, Router Conn, and Graphics Conn—by exchanging requests and replies. The lines connecting the component and connector elements in the architecture represent the elements' interaction paths.

As-Designed vs. As-Implemented Architecture



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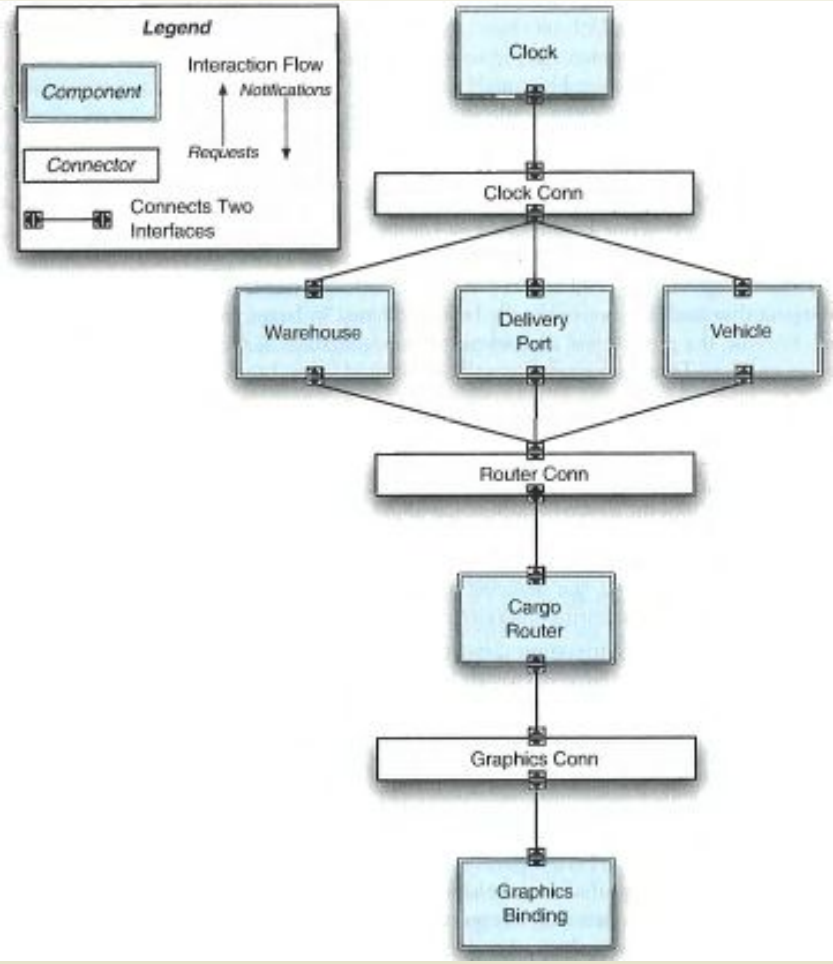
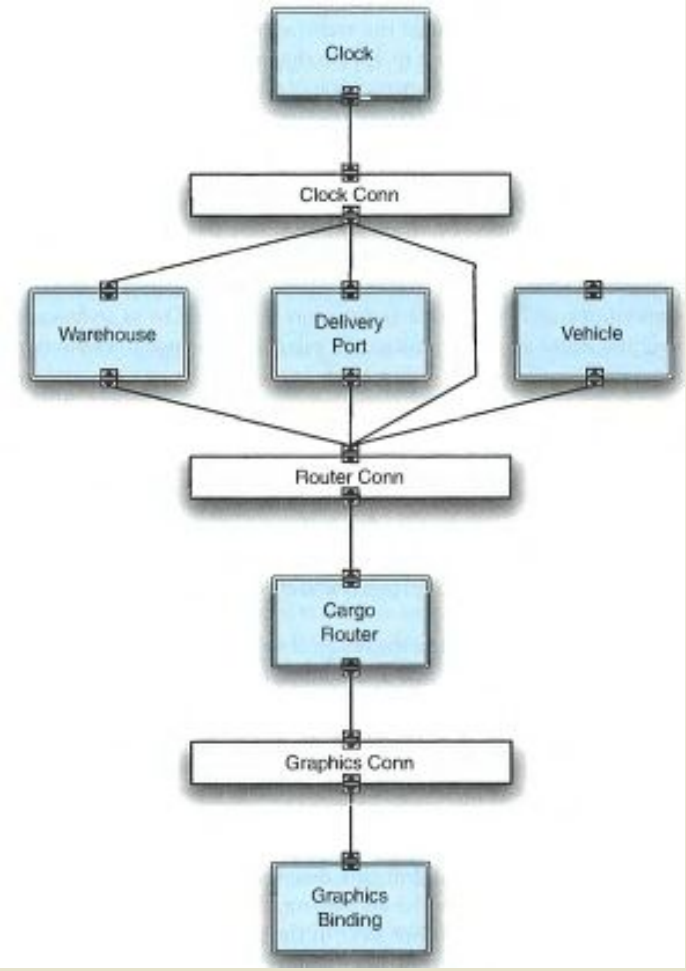
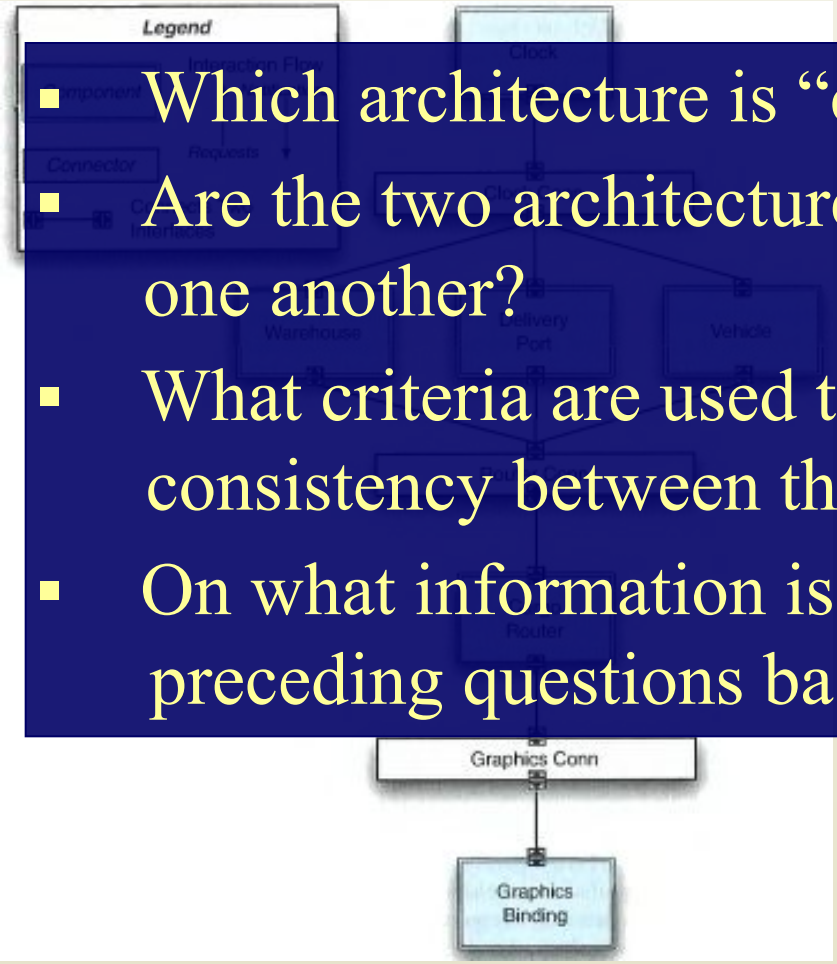
As-Designed vs. As-Implemented Architecture

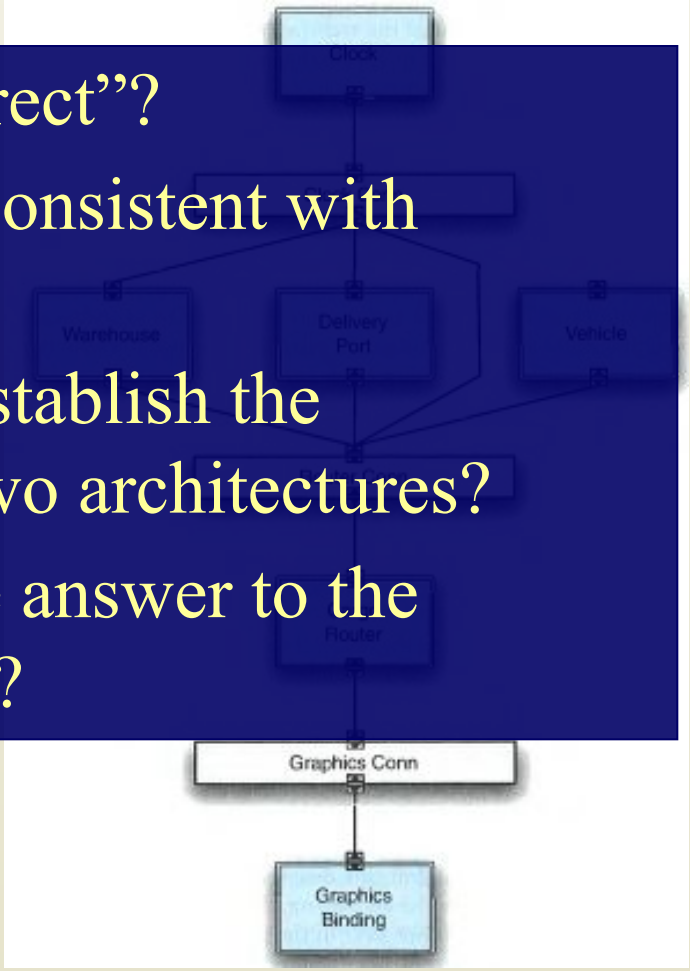




As-Designed vs. As-Implemented Architecture

- Which architecture is "correct"?
- Are the two architectures consistent with one another?
- What criteria are used to establish the consistency between the two architectures?
- On what information is the answer to the preceding questions based?





Architectural Evolution

- When a system evolves, ideally its prescriptive architecture is modified first
- In practice, the system – and thus its descriptive architecture – is often directly modified
- This happens because of
  - Developer sloppiness
  - Perception of short deadlines which prevent thinking through and documenting
  - Lack of documented prescriptive architecture
  - Need or desire for code optimizations
  - Inadequate techniques or tool support

Architectural Degradation

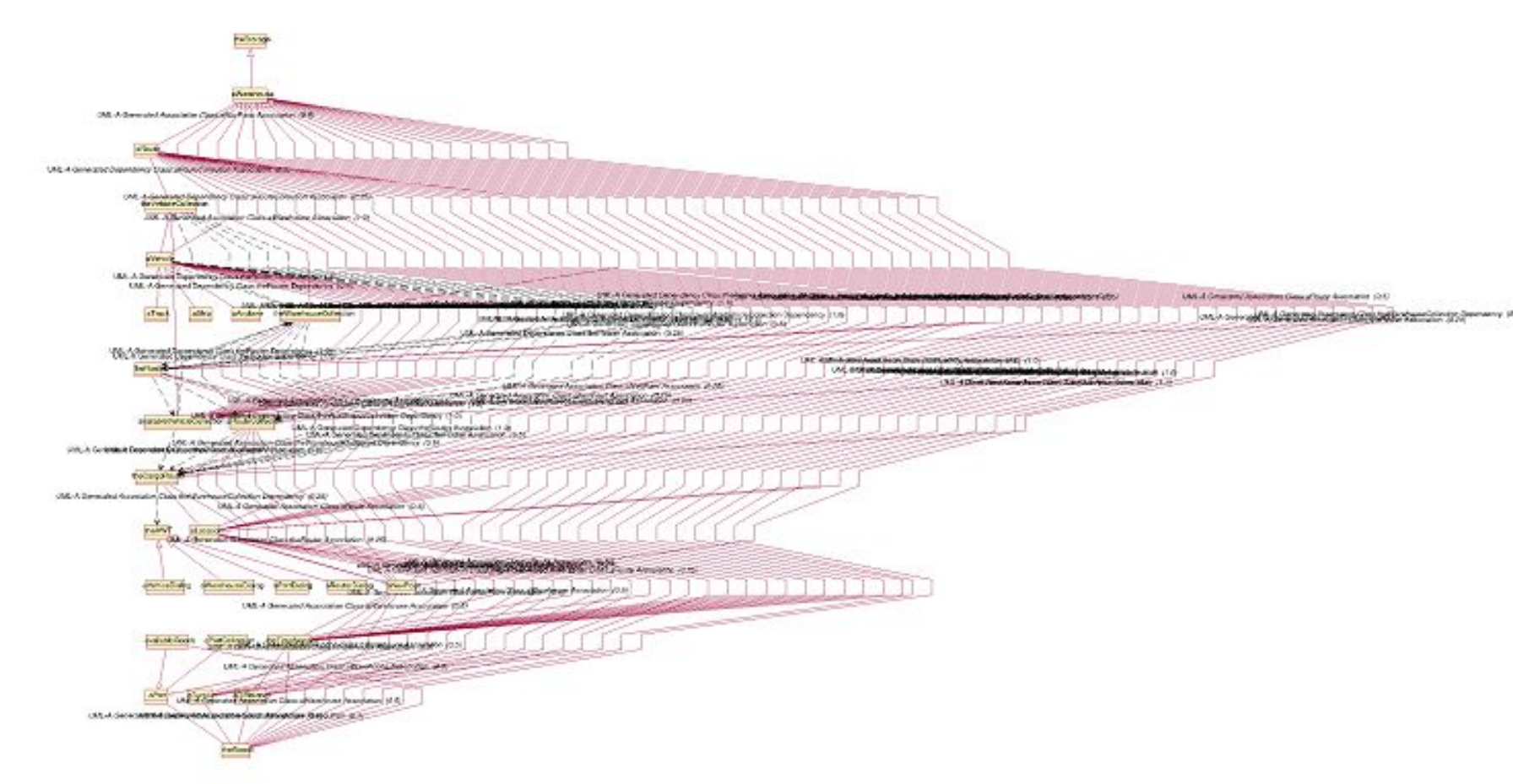
- Two related concepts
  - Architectural drift
  - Architectural erosion
- *Architectural drift* is introduction of principal design decisions into a system's descriptive architecture that
  - are not included in, encompassed by, or implied by the prescriptive architecture
  - but which do not violate any of the prescriptive architecture's design decisions
- *Architectural erosion* is the introduction of architectural design decisions into a system's descriptive architecture that violate its prescriptive architecture

In the cargo routing application example, the removal of the link between the Vehicle component and its adjacent connector in the descriptive architecture would have resulted in architectural erosion had it not been justified and had the prescriptive architecture not been properly updated. The removal of this link introduces the potential danger that the vehicles controlled by the system are unable to synchronize properly with the rest of the system and deliver their cargo on time. In this case, the system developers realized that Vehicles do not need to keep track of time internally so long as the Cargo Router component gives them appropriate instructions. They discussed this observation with the architects, who agreed and updated the prescriptive architecture accordingly.

Architectural Recovery

- If architectural degradation is allowed to occur, one will be forced to *recover* the system's architecture sooner or later
- *Architectural recovery* is the process of determining a software system's architecture from its implementation-level artifacts
- Implementation-level artifacts can be
  - Source code
  - Executable files
  - Java .class files

Implementation-Level View of an Application



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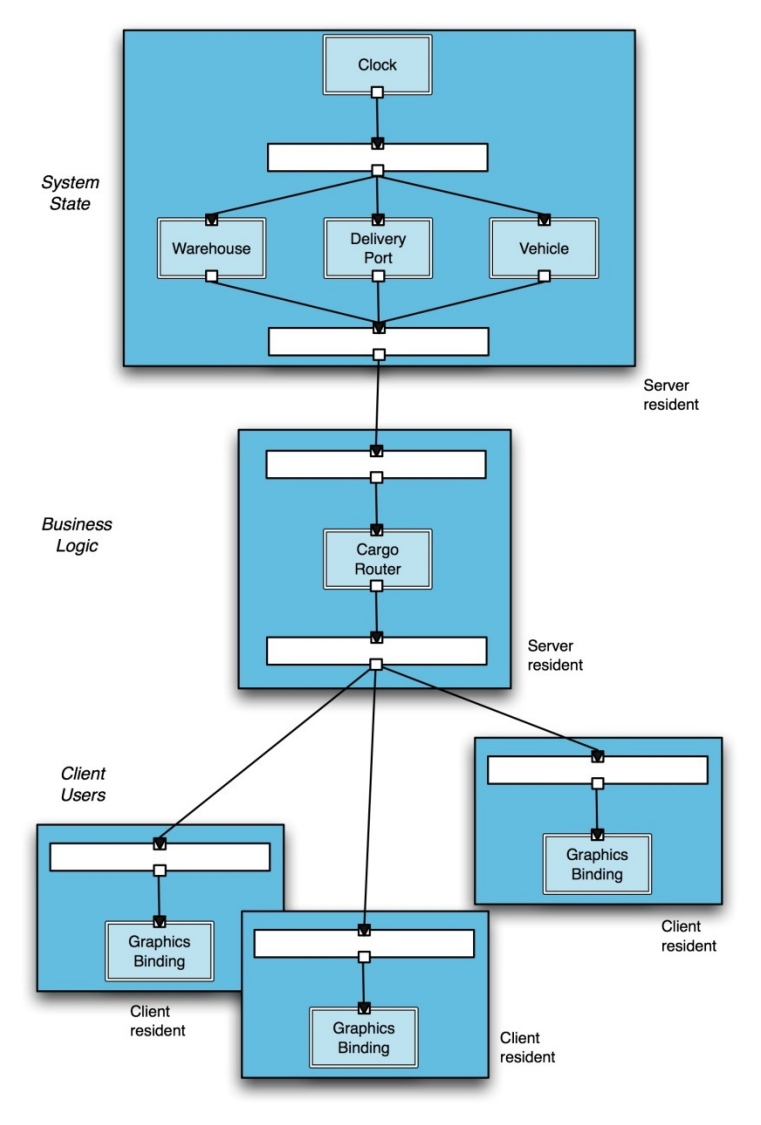
Deployment

- A software system cannot fulfill its purpose until it is *deployed*
  - Executable modules are physically placed on the hardware devices on which they are supposed to run
- The deployment view of an architecture can be critical in assessing whether the system will be able to satisfy its requirements
- Possible assessment dimensions
  - Available memory
  - Power consumption
  - Required network bandwidth

requirements. For example, placing many large components on a small device with limited memory and CPU power, or transferring high volumes of data over a network link with low bandwidth will negatively impact the system, much like incorrectly implementing its functionality will. An example deployment perspective on the cargo routing application's

A System's Deployment Architectural Perspective



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Software Architecture's Elements

- A software system's architecture typically is not (and should not be) a uniform monolith
- A software system's architecture should be a composition and interplay of different elements
  - Processing
  - Data, also referred as information or state
  - Interaction

Components

- Elements that encapsulate processing and data in a system's architecture are referred to as *software components*
- **Definition**
  - A *software component* is an architectural entity that
    - encapsulates a subset of the system's functionality and/or data
    - restricts access to that subset via an explicitly defined interface
    - has explicitly defined dependencies on its required execution context
- Components typically provide application-specific services

Connectors

- In complex systems *interaction* may become more important and challenging than the functionality of the individual components
- **Definition**
  - A *software connector* is an architectural building block tasked with effecting and regulating interactions among components
- In many software systems connectors are usually simple procedure calls or shared data accesses
  - Much more sophisticated and complex connectors are possible!
- Connectors typically provide application-independent interaction facilities

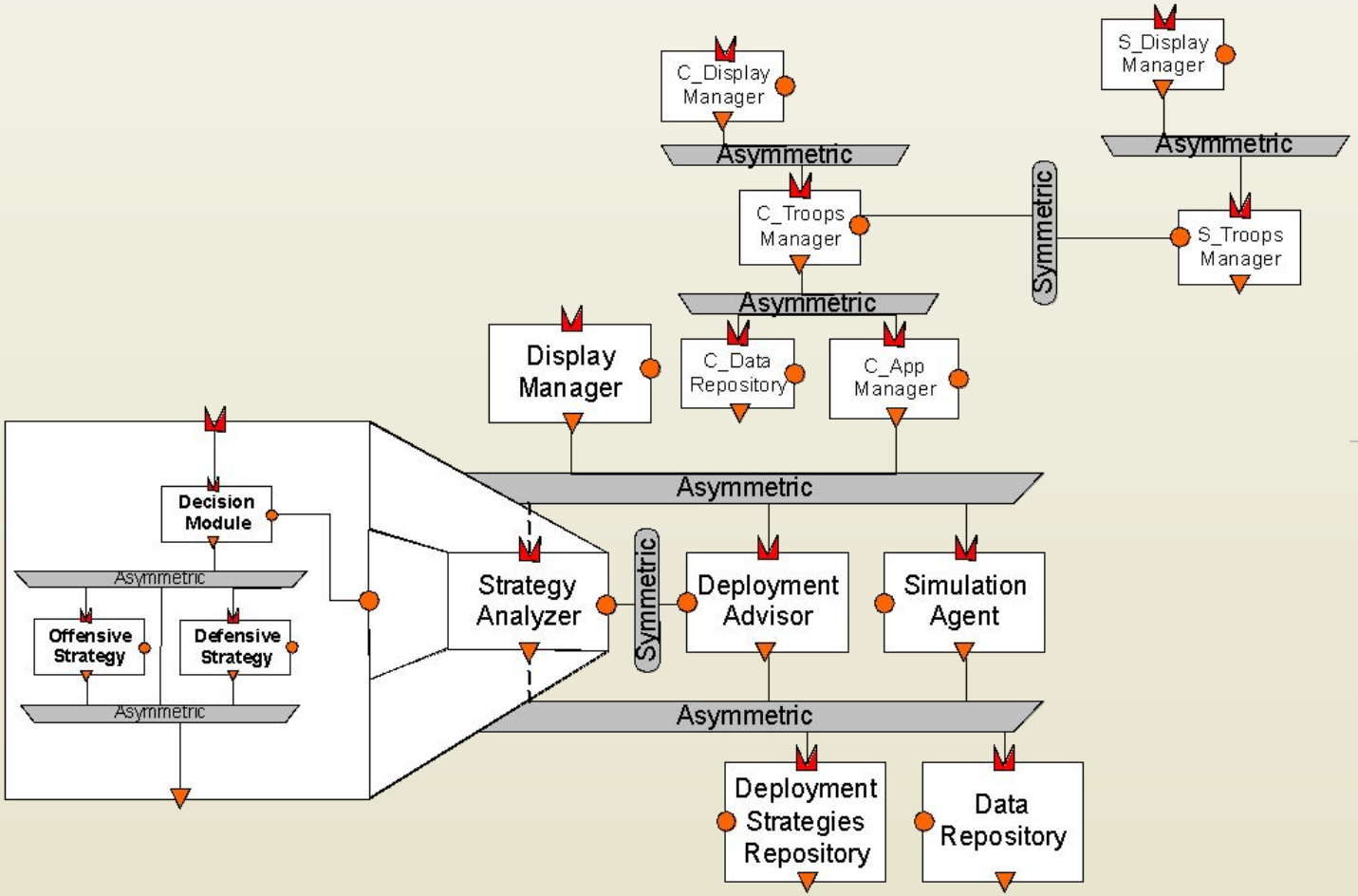
Examples of Connectors

- Procedure call connectors
- Shared memory connectors
- Message passing connectors
- Streaming connectors
- Distribution connectors
- Wrapper/adaptor connectors

Configurations

- Components and connectors are composed in a specific way in a given system's architecture to accomplish that system's objective
- **Definition**
  - An *architectural configuration*, or topology, is a set of specific associations between the components and connectors of a software system's architecture

An Example Configuration



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Architectural Styles

- Certain design choices regularly result in solutions with superior properties
  - Compared to other possible alternatives, solutions such as this are more elegant, effective, efficient, dependable, evolvable, scalable, and so on
- **Definition**
  - An *architectural style* is a named collection of architectural design decisions that
    - are applicable in a given development context
    - constrain architectural design decisions that are specific to a particular system within that context
    - elicit beneficial qualities in each resulting system

Architectural Patterns

- **Definition**
  - An *architectural pattern* is a set of architectural design decisions that are applicable to a recurring design problem, and parameterized to account for different software development contexts in which that problem appears
- A widely used pattern in modern distributed systems is the *three-tiered system* pattern
  - Science
  - Banking
  - E-commerce
  - Reservation systems

Three-Tiered Pattern

- Front Tier
  - Contains the user interface functionality to access the system's services
- Middle Tier
  - Contains the application's major functionality
- Back Tier
  - Contains the application's data access and storage functionality

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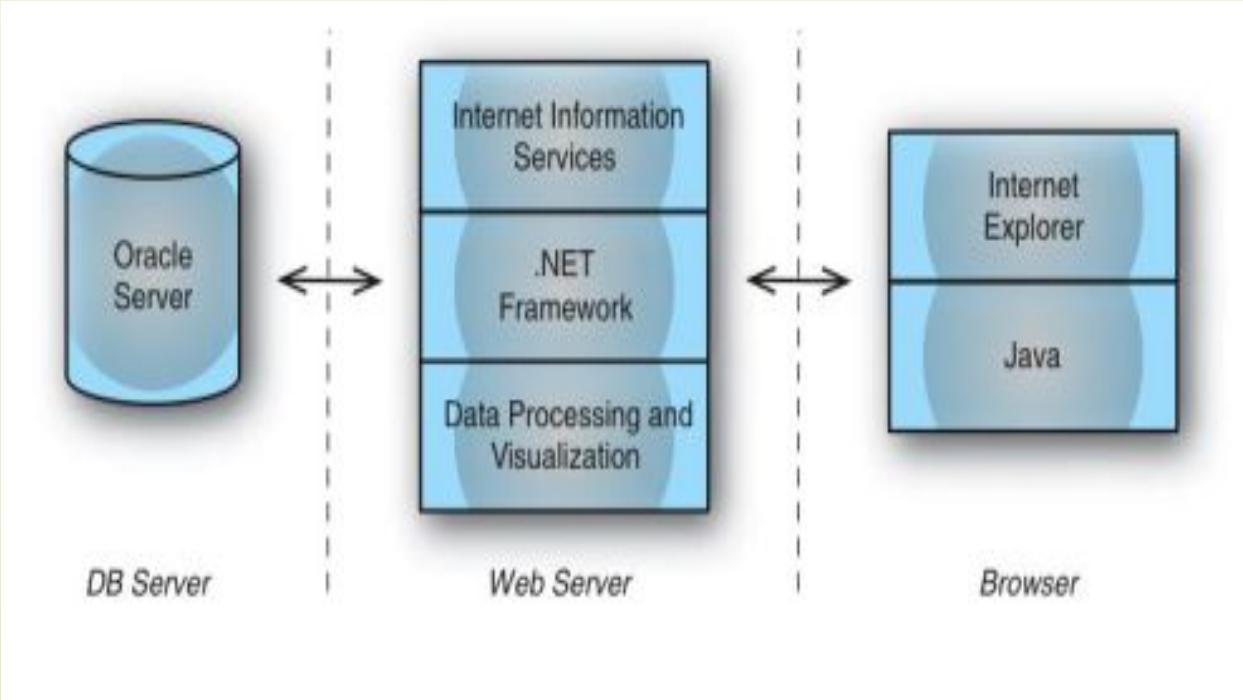
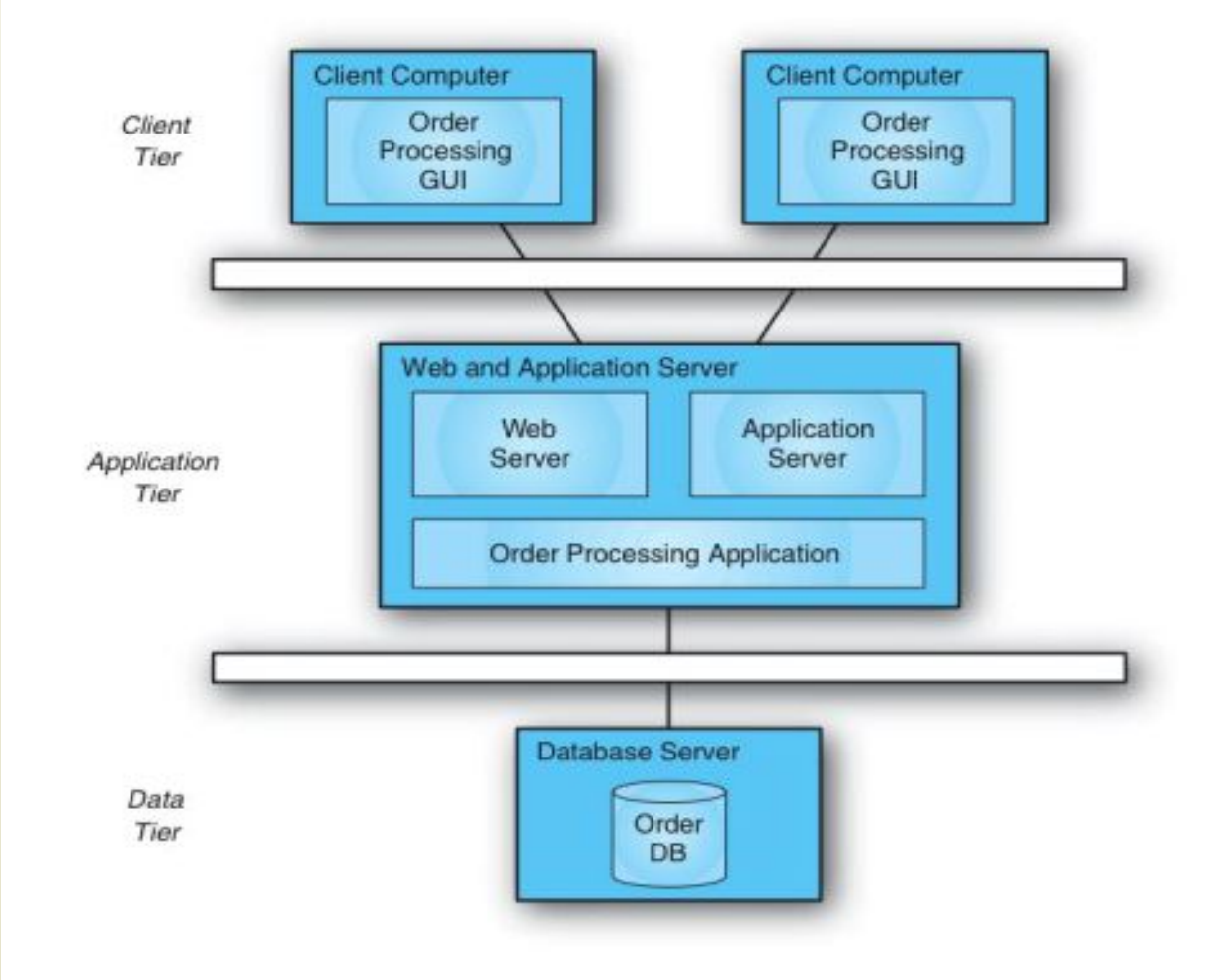
Architectural Models



Architectural Models



Architectural Models, Views, and Visualizations

- Architecture Model
  - An artifact documenting some or all of the architectural design decisions about a system
- Architecture Visualization
  - A way of depicting some or all of the architectural design decisions about a system to a stakeholder
- Architecture View
  - A subset of related architectural design decisions

Architectural Processes

- Architectural design
- Architecture modeling and visualization
- Architecture-driven system analysis
- Architecture-driven system implementation
- Architecture-driven system deployment, runtime redeployment, and mobility
- Architecture-based design for non-functional properties, including security and trust
- architectural adaptation

Stakeholders in a System's Architecture

- Architects
- Developers
- Testers
- Managers
- Customers
- Users
- Vendors